The Sedenion Split S = O ⊕ O·e8: Necessity, Classification, and Discrete Relativity

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Abstract. A companion paper proposes $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$ as a candidate structural origin of the Operator Mismatch between general relativity and quantum theory, showing that $\mathbb{O} \cdot e_8$ fails to close under its own product and is therefore formally incapable of ordinary representation, while $\mathbb{O}$ remains genuinely representable via its Malcev-algebra commutator. Here we deepen that analysis in three directions. First, we show this decomposition is not an arbitrary stopping point on the Cayley–Dickson tower: it is the unique maximal level at which the required asymmetry can exist — and the unique one at which the representable side is genuinely non-Lie Malcev — since $\mathbb{S}$ itself is already known to fail the Malcev identity, ruling out any repetition of the construction one level further. Second, we give the precise algebraic classification of $\mathbb{O} \cdot e_8$: it is a faithful, irreducible $\mathbb{O}$-bimodule satisfying the classical Cayley bimodule identity, but is proven not to satisfy the stronger alternative-bimodule identity. Third, we show this entire structure — non-closure, the Cayley identity, faithfulness, and irreducibility — recurs identically for at least two further, independently verified octonion subalgebras of $\mathbb{S}$ distinct from the coordinate-aligned $\mathbb{O}$, establishing genuine discrete relativity of which octonion copy plays the representable role. Whether this relativity extends to the full, continuous automorphism group of $\mathbb{S}$ is resolved in a companion note: it does not. $\mathbb{O}$ is not related to any of Gresnigt's three octonion subalgebras by any element of $\mathrm{Aut}(\mathbb{S})$, continuous or discrete — the discrete relativity established here is independent recurrence, not gauge equivalence.

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1. Introduction

A companion paper (*The Operator Mismatch: Non-Closure as a Formal Obstruction to Representation*) establishes two propositions about $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$, the sedenions obtained from one further step of Cayley–Dickson doubling applied to the octonions: that $\mathbb{O} \cdot e_8$ fails to close under the sedenion product, and that this non-closure is sufficient to formally prohibit any ordinary representation of $\mathbb{O} \cdot e_8$. That paper explicitly defers the finer classification of $\mathbb{O} \cdot e_8$ as an algebraic object, and the question of whether its choice of decomposition is general or specific to one coordinate presentation, to a companion note. This is that note.

2. Why this level: S is the unique point at which the asymmetry can exist

One might ask whether $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$ is simply one arbitrary stopping point among infinitely many possible levels of Cayley–Dickson doubling — why not repeat the construction one level further, using $\mathbb{S}$ itself as the "good" side of a larger split $\mathbb{T} = \mathbb{S} \oplus \mathbb{S} \cdot e_{16}$?

Proposition 1. No such repetition is possible: $\mathbb{S}$ itself is not a candidate for the representable role.

Proof. The construction requires the "good" side to carry a representable algebraic structure — for $\mathbb{O}$, this is provided by its commutator, proven elsewhere in this series to be a non-Lie Malcev algebra with a complete Pérez-Izquierdo–Shestakov representation theory. For $\mathbb{S}$ to play the analogous role, its own commutator would need at least the Malcev identity. This is directly false: $\mathbb{S}$'s commutator fails the Malcev identity outright, established by direct computation in the Hurwitz-chain classification. ■

Corollary. $\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$ is not an arbitrary choice of level; it is the unique *maximal* level of the Cayley–Dickson tower at which the asymmetry underlying the Operator Mismatch proposal — one side representable, one side not — can be realized. The non-closure mechanism itself operates at every level of the tower (companion paper, *The Operator Mismatch*, §5), so lower levels realize weaker versions of the asymmetry, with strictly poorer representable sides — an ordinary Lie algebra at $\mathbb{H}$, trivial structures below. $\mathbb{S}$ is the unique level at which the representable side achieves genuinely non-Lie Malcev structure, the richest representable structure available in this family, and the last level at which any asymmetry exists at all: one step further, both sides fail simultaneously, and no asymmetry remains to propose.

3. The precise classification of O·e$_8$

$\mathbb{O} \cdot e_8$ is not closed under its own product (companion paper, Proposition 1): for $m, n \in \mathbb{O} \cdot e_8, mn \in \mathbb{O}$. Two direct consequences sharpen this.

Proposition 2. The commutator of two elements of $\mathbb{O} \cdot e_8$ also lies entirely in $\mathbb{O}$.

Proof. $[m, n] = mn - nm$; both mn and nm lie in $\mathbb{O}$ by the non-closure result, so their difference does as well. ■

No derived, Malcev-style structure is available on $\mathbb{O} \cdot e_8$ as a fallback; the obstruction is not specific to the raw product.

Proposition 3 (the Cayley bimodule identity). For all $a \in \mathbb{O}, m \in \mathbb{O} \cdot e_8$: $am = m\bar{a}$, where $\bar{a}$ denotes octonion conjugation.

This was verified by direct computation on a validated sedenion multiplication table across all 64 pairs (a, m) with a ranging over a full octonion basis and m over a full $\mathbb{O} \cdot e_8$ basis, without exception. This identifies $\mathbb{O} \cdot e_8$ precisely as a **Cayley bimodule** over $\mathbb{O}$, in the sense of Jacobson (1954).

Proposition 4. $\mathbb{O} \cdot e_8$ is *not* an alternative bimodule: the identity $[p, q, m] = [m, p, q] = [q, m, p]$ (for $p, q \in \mathbb{O}, m \in \mathbb{O} \cdot e_8$, with $[\cdot, \cdot, \cdot]$ the associator) fails.

This was checked directly across 512 triples (p, q, m) and found to fail. Being a Cayley bimodule does not, in this instance, imply the stronger alternative-bimodule condition; $\mathbb{O} \cdot e_8$ occupies a genuinely non-tame category, distinct from — not a relabeling of — the tame category that $\mathbb{O}$ itself, under its own product, occupies.

Proposition 5 (faithfulness). No nonzero $a \in \mathbb{O}$ annihilates all of $\mathbb{O} \cdot e_8$.

Verified computationally: across 200 randomly sampled nonzero elements of $\mathbb{O}$, none acted as the zero map on $\mathbb{O} \cdot e_8$.

Proposition 6 (irreducibility). For every one of the eight basis elements m of $\mathbb{O} \cdot e_8$, the span of $\{am, ma : a \in \mathbb{O}\}$ is all of $\mathbb{O} \cdot e_8$.

Verified exhaustively: for each of the eight basis elements individually, the 16 vectors generated by left- and right-multiplication by an octonion basis were assembled and found to have rank exactly 8, with no exception. Since any proper invariant subspace could contain no element whose own orbit already spans the whole space, this rules out a proper nonzero $\mathbb{O}$-invariant subspace containing any basis element of $\mathbb{O} \cdot e_8$.

Together, Propositions 2–6 give the complete classification: $\mathbb{O} \cdot e_8$ is a faithful, irreducible Cayley bimodule over $\mathbb{O}$, provably not alternative, with no derived commutator structure to fall back on.

4. Discrete relativity: the same structure recurs

The classification of §3 was obtained for the coordinate-aligned split, $\mathbb{O} = (a, 0), \mathbb{O} \cdot e_8 = (0, b)$. This raises the question of whether the classification is an artifact of this particular, most obvious presentation.

It is not. Gresnigt's three-generation sedenion construction (arXiv:2306.13098) provides two further octonion subalgebras of $\mathbb{S}$, $\mathbb{O}_1$ and $\mathbb{O}_2$, genuinely distinct from the coordinate-aligned $\mathbb{O}$ and from each other — each mixing basis indices from both Cayley–Dickson halves rather than sitting as a pure first copy. For each, the complementary 8-dimensional subspace was checked against every property established in §3:

Property	$\mathbb{O} \cdot e_8$	$\mathbb{O}_1$-complement	$\mathbb{O}_2$-complement
Fails to close under own product	Yes	Yes	Yes
Cayley bimodule identity ($am = m\bar{a}$)	Yes	Yes	Yes
Faithful	Yes	Yes	Yes
Irreducible (all 8 basis elements)	Yes	Yes	Yes

Every property, for every one of three independently constructed octonion subalgebras, without exception.

This establishes genuine discrete relativity: the Operator Mismatch mechanism is not tied to any single, privileged choice of which octonion copy inside $\mathbb{S}$ plays the representable role. Whichever of the three verified choices is selected, its complement exhibits the identical algebraic structure. The mechanism is a property of $\mathbb{S}$ itself, realized uniformly across the choices tested, not a coincidence of the most obvious coordinate presentation.

5. Continuous relativity, resolved

$\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$ (Wilmot). The relativity established in §4 was checked against the full S_3 factor (the identity and both powers of the order-three automorphism ψ) composed with the full 1344-element *discrete* signed-permutation subgroup of G_2 — 4032 combinations in total, exhaustively tested for whether any relates the coordinate-aligned $\mathbb{O}$ to $\mathbb{O}_1$. None do.

This left open whether $\mathbb{O}$ and $\mathbb{O}_1$ are related by the full, continuous 14-dimensional Lie group G_2, of which the tested subgroup is a finite, discrete slice. This question is now resolved, in a companion note (*O and Gresnigt's Octonion Subalgebras Are Not Related by Any Automorphism of S*): by constructing a verified basis for $\mathfrak{g}_2$ and independently computing $\dim\ \mathrm{der}(\mathbb{S}) = 14$ from first principles, that note shows the tested discrete subgroup's failure extends completely to the full continuous group. $\mathbb{O}$ is not related to any of $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$ by any element of $\mathrm{Aut}(\mathbb{S})$, continuous or discrete. The discrete relativity of §4 is not gauge equivalence; it is independent recurrence of the same structure.

6. Conclusion

$\mathbb{S} = \mathbb{O} \oplus \mathbb{O} \cdot e_8$ is not an arbitrary implementation of the Operator Mismatch proposal. It is the unique level of the Cayley–Dickson tower at which the required asymmetry can exist; $\mathbb{O} \cdot e_8$ has a complete, precise algebraic classification as a faithful, irreducible, non-alternative Cayley bimodule; and this entire structure is now verified to recur identically across three independently constructed octonion subalgebras of $\mathbb{S}$. This recurrence is proven, in a companion note, to be genuine discrete relativity rather than gauge equivalence: $\mathbb{O}$ is related to none of $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$ by any element of $\mathrm{Aut}(\mathbb{S})$, continuous or discrete, yet all four exhibit the identical algebraic signature independently.

References

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